

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	30.1	10	0.4	0.2	0.1	59.2
U3 - Toxicity prior Beta(0.15, 0.85)	31.2	9.3	0.1	0.2	0	59.2
U2 - Toxicity prior Beta(0.3, 0.7)	28.6	12.3	0.3	0.3	0.1	58.4
U3 - Toxicity prior Beta(0.3, 0.7)	31	9.4	0.5	0.3	0.1	58.7
U2 - Toxicity prior Beta(0.5, 0.5)	30.3	11.2	1.5	0.4	0.1	56.5
U3 - Toxicity prior Beta(0.5, 0.5)	33.1	10.2	0.6	0.2	0.1	55.8
U2 - Toxicity prior Beta(1, 1)	31.5	10.6	1.2	0.5	0.1	56.1
U3 - Toxicity prior Beta(1, 1)	33.8	8.6	0.6	0.2	0.1	56.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	15.8	3.9	1.8	0.6	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	15.9	3.8	1.8	0.6	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	15.7	3.9	1.8	0.7	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	15.6	4	1.8	0.7	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	15.3	4.4	1.9	0.8	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	15.3	4.3	2	0.8	0.3	
U2 - Toxicity prior Beta(1, 1)	15.3	4.3	2	0.8	0.3	
U3 - Toxicity prior Beta(1, 1)	15.4	4.3	1.9	0.8	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2	0.5	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2	0.5	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	2	0.5	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	2	0.5	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2	0.6	0.3	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2	0.6	0.3	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	1.9	0.7	0.3	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	2	0.6	0.3	0.1	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	35.2	19.3	31.9	9.5	2.2	1.9
U3 - Toxicity prior Beta(0.15, 0.85)	39.3	19	29.9	8.4	1.6	1.8
U2 - Toxicity prior Beta(0.3, 0.7)	34.1	18.6	32.3	10.3	2.9	1.8
U3 - Toxicity prior Beta(0.3, 0.7)	38.2	18	29.6	10.6	1.8	1.8
U2 - Toxicity prior Beta(0.5, 0.5)	32.7	15.3	33.4	13.2	3.6	1.8
U3 - Toxicity prior Beta(0.5, 0.5)	36.4	17.3	30.4	12	2.1	1.8
U2 - Toxicity prior Beta(1, 1)	32.4	15.3	34.7	12.5	3.3	1.8
U3 - Toxicity prior Beta(1, 1)	35.3	17.7	30.9	12.2	2.1	1.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.9	5.4	7.3	3.7	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	11.8	5.5	7.2	3.7	1.6	
U2 - Toxicity prior Beta(0.3, 0.7)	11.6	5.4	7.1	3.8	1.8	
U3 - Toxicity prior Beta(0.3, 0.7)	11.7	5.2	7.2	3.8	1.8	
U2 - Toxicity prior Beta(0.5, 0.5)	11	5.1	7.2	4.3	2	
U3 - Toxicity prior Beta(0.5, 0.5)	11	5.1	7.3	4.3	2.1	
U2 - Toxicity prior Beta(1, 1)	11.1	5	7.3	4.3	2	
U3 - Toxicity prior Beta(1, 1)	11	5	7.3	4.4	2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.4	1.3	0.8	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.5	1.3	0.9	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.4	1.3	0.9	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.4	1.3	0.9	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.3	1.4	0.9	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.3	1.4	0.9	0.3	
U2 - Toxicity prior Beta(1, 1)	0.3	2.3	1.4	0.9	0.3	
U3 - Toxicity prior Beta(1, 1)	0.4	2.3	1.4	0.9	0.3	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.4	43.8	19.3	6.8	1.5	10.2
U3 - Toxicity prior Beta(0.15, 0.85)	19.8	47	16.5	5.2	1.3	10.2
U2 - Toxicity prior Beta(0.3, 0.7)	17.5	43.3	18.5	8.9	1.6	10.2
U3 - Toxicity prior Beta(0.3, 0.7)	19.4	45.9	16.8	6.8	1.1	10
U2 - Toxicity prior Beta(0.5, 0.5)	17	39.6	21.1	10.2	2.4	9.7
U3 - Toxicity prior Beta(0.5, 0.5)	19.1	43.7	18.4	7.4	1.6	9.8
U2 - Toxicity prior Beta(1, 1)	17.2	38.2	21.9	11	2	9.7
U3 - Toxicity prior Beta(1, 1)	19.2	43.5	18.2	7.9	1.5	9.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.1	7.9	5.9	3.4	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	10.1	7.9	6	3.4	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	9.9	7.7	5.9	3.5	1.7	
U3 - Toxicity prior Beta(0.3, 0.7)	9.9	7.7	5.9	3.5	1.7	
U2 - Toxicity prior Beta(0.5, 0.5)	9.5	7.5	6.1	4	1.9	
U3 - Toxicity prior Beta(0.5, 0.5)	9.5	7.4	6.2	3.9	1.9	
U2 - Toxicity prior Beta(1, 1)	9.5	7.3	6.2	4	1.9	
U3 - Toxicity prior Beta(1, 1)	9.5	7.3	6.3	4	1.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3	1.2	1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3	1.2	1	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.9	1.2	1	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.9	1.2	1	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.3	1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.4	1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.2	2.8	1.3	1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	1.1	0.3	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	7.9	8.5	23.2	41.6	16.8	2
U3 - Toxicity prior Beta(0.15, 0.85)	12.9	9.6	20.9	40.6	14.1	1.9
U2 - Toxicity prior Beta(0.3, 0.7)	7.8	9.4	21	42.8	17.1	1.9
U3 - Toxicity prior Beta(0.3, 0.7)	13	10.1	19.1	40.6	15.3	1.9
U2 - Toxicity prior Beta(0.5, 0.5)	6.1	7.2	19.2	46.8	18.8	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	10.1	8.7	17.7	46	15.6	1.9
U2 - Toxicity prior Beta(1, 1)	5.4	7.3	19.8	46.2	19.4	1.9
U3 - Toxicity prior Beta(1, 1)	9.7	9	18.1	44	17.3	1.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.7	4.4	6.8	8	5.8	
U3 - Toxicity prior Beta(0.15, 0.85)	4.7	4.4	6.8	8.1	5.8	
U2 - Toxicity prior Beta(0.3, 0.7)	4.6	4.3	6.5	8.3	6.1	
U3 - Toxicity prior Beta(0.3, 0.7)	4.5	4.3	6.5	8.2	6.2	
U2 - Toxicity prior Beta(0.5, 0.5)	4.3	3.9	6	8.9	6.5	
U3 - Toxicity prior Beta(0.5, 0.5)	4.3	4	6.1	8.9	6.5	
U2 - Toxicity prior Beta(1, 1)	4.3	4	6	9	6.5	
U3 - Toxicity prior Beta(1, 1)	4.3	3.9	6	9	6.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	1.5	2.2	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	1.4	2.2	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	1.4	2.2	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.8	1.4	2.2	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.5	2.2	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.5	2.2	0.7	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.5	2.2	0.8	
U3 - Toxicity prior Beta(1, 1)	0.2	2.7	1.5	2.2	0.8	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	4.7	34.9	26.3	20.4	9.5	4.2
U3 - Toxicity prior Beta(0.15, 0.85)	7.5	41.5	24.6	15.6	6.6	4.2
U2 - Toxicity prior Beta(0.3, 0.7)	4.8	33.9	26.5	20.4	10.2	4.2
U3 - Toxicity prior Beta(0.3, 0.7)	7.3	40.9	24.5	16.7	6.4	4.2
U2 - Toxicity prior Beta(0.5, 0.5)	4.5	31.8	28.4	20.1	11	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	7.6	39.5	25.2	17.1	6.4	4.2
U2 - Toxicity prior Beta(1, 1)	4.5	32.6	26.4	21	11.3	4.2
U3 - Toxicity prior Beta(1, 1)	7.9	40.2	24.5	16.8	6.4	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5	6.5	6.9	5.8	5.1	
U3 - Toxicity prior Beta(0.15, 0.85)	5	6.6	6.7	5.9	5.2	
U2 - Toxicity prior Beta(0.3, 0.7)	5	6.5	6.6	5.9	5.4	
U3 - Toxicity prior Beta(0.3, 0.7)	5	6.5	6.6	5.9	5.4	
U2 - Toxicity prior Beta(0.5, 0.5)	4.9	6.1	6.8	6	5.7	
U3 - Toxicity prior Beta(0.5, 0.5)	4.9	6.2	6.6	6.1	5.7	
U2 - Toxicity prior Beta(1, 1)	4.9	6.1	6.7	6.1	5.7	
U3 - Toxicity prior Beta(1, 1)	4.9	6.1	6.7	6	5.7	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	3.4	1.7	2.1	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	3.4	1.7	2.1	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	3.3	1.7	2.1	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	3.3	1.7	2.1	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	3.2	1.8	2.1	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	3.2	1.8	2.2	1.1	
U2 - Toxicity prior Beta(1, 1)	0.1	3.3	1.8	2.1	1.1	
U3 - Toxicity prior Beta(1, 1)	0.2	3.2	1.8	2.1	1.1	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	69.3	12.7	2.8	0.8	0.3	14.1
U3 - Toxicity prior Beta(0.15, 0.85)	70.1	12.3	2.4	0.8	0.2	14.2
U2 - Toxicity prior Beta(0.3, 0.7)	68.4	13.7	2.5	1	0.3	14.1
U3 - Toxicity prior Beta(0.3, 0.7)	69.5	13.3	2.3	0.8	0.2	13.9
U2 - Toxicity prior Beta(0.5, 0.5)	66.6	13.8	4.2	1.3	0.4	13.7
U3 - Toxicity prior Beta(0.5, 0.5)	68.6	12.9	3.7	1	0.3	13.5
U2 - Toxicity prior Beta(1, 1)	67.1	13.2	4.2	1.4	0.3	13.8
U3 - Toxicity prior Beta(1, 1)	69.2	12.3	3.5	0.8	0.3	13.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	20.1	4	2.1	0.9	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	20.1	4	2.2	0.9	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	20.1	4	2.2	0.9	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	20.1	4	2.2	0.9	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	19.4	4.3	2.3	1.2	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	19.5	4.3	2.3	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	19.4	4.2	2.3	1.2	0.3	
U3 - Toxicity prior Beta(1, 1)	19.4	4.3	2.3	1.1	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	1.5	0.4	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	1.5	0.5	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	1.5	0.5	0.3	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	1.5	0.5	0.3	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	1.5	0.5	0.3	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.5	0.5	0.3	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	1.5	0.5	0.3	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	1.5	0.5	0.3	0.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	26.2	38.5	20.1	9.8	3.1	2.3
U3 - Toxicity prior Beta(0.15, 0.85)	28.5	42	18	7.6	1.7	2.2
U2 - Toxicity prior Beta(0.3, 0.7)	26.3	37.6	20.9	9.6	3.4	2.2
U3 - Toxicity prior Beta(0.3, 0.7)	28.3	42.2	17.4	7.8	2.1	2.2
U2 - Toxicity prior Beta(0.5, 0.5)	25.3	34.2	22.8	11.1	4.4	2.2
U3 - Toxicity prior Beta(0.5, 0.5)	27.9	37.6	19.8	9.9	2.6	2.2
U2 - Toxicity prior Beta(1, 1)	25.7	33.1	22.6	12	4.4	2.2
U3 - Toxicity prior Beta(1, 1)	27.6	37.8	19.4	10.3	2.7	2.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.1	7.3	5.6	3.6	2	
U3 - Toxicity prior Beta(0.15, 0.85)	11.1	7.3	5.6	3.6	2	
U2 - Toxicity prior Beta(0.3, 0.7)	11.1	7.1	5.6	3.7	2.2	
U3 - Toxicity prior Beta(0.3, 0.7)	11.1	7.1	5.5	3.6	2.3	
U2 - Toxicity prior Beta(0.5, 0.5)	10.5	6.9	5.7	4	2.5	
U3 - Toxicity prior Beta(0.5, 0.5)	10.5	6.9	5.7	4	2.5	
U2 - Toxicity prior Beta(1, 1)	10.5	6.8	5.7	4	2.5	
U3 - Toxicity prior Beta(1, 1)	10.5	6.9	5.8	4	2.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.6	1	1	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	1	0.9	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1	1	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.6	1	1	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.5	1.1	1	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.5	1.1	1	0.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.5	1.1	1	0.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.5	1.1	1	0.4	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	43.2	8.4	1.7	0.6	0.1	46
U3 - Toxicity prior Beta(0.15, 0.85)	44.5	8.2	1.1	0.4	0.1	45.7
U2 - Toxicity prior Beta(0.3, 0.7)	42.5	9.3	2	0.5	0.3	45.4
U3 - Toxicity prior Beta(0.3, 0.7)	45.2	8	1.1	0.6	0.2	44.9
U2 - Toxicity prior Beta(0.5, 0.5)	42	9.6	2.6	0.9	0.5	44.4
U3 - Toxicity prior Beta(0.5, 0.5)	45	8	1.7	0.8	0.3	44.2
U2 - Toxicity prior Beta(1, 1)	43	8.6	2.6	1	0.5	44.3
U3 - Toxicity prior Beta(1, 1)	44.8	7.9	2.1	0.6	0.4	44.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	18	3.5	1.6	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	18	3.6	1.5	0.4	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	17.9	3.5	1.6	0.5	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	17.9	3.5	1.6	0.5	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	17.4	3.9	1.8	0.7	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	17.4	3.9	1.7	0.7	0.2	
U2 - Toxicity prior Beta(1, 1)	17.3	3.9	1.8	0.7	0.2	
U3 - Toxicity prior Beta(1, 1)	17.3	3.9	1.8	0.7	0.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	1.6	0.4	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	1.6	0.4	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	1.6	0.4	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	1.5	0.4	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	1.5	0.5	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.5	0.5	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.4	1.5	0.5	0.2	0	
U3 - Toxicity prior Beta(1, 1)	0.4	1.5	0.5	0.2	0	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	15.9	12.7	17.1	13.6	21.2	19.5
U3 - Toxicity prior Beta(0.15, 0.85)	21.7	13.4	14.9	12.4	18.2	19.4
U2 - Toxicity prior Beta(0.3, 0.7)	14.4	13.4	17	13.5	22.5	19.2
U3 - Toxicity prior Beta(0.3, 0.7)	19.9	13.5	15.5	12.2	19.7	19.2
U2 - Toxicity prior Beta(0.5, 0.5)	13.7	10.5	18	13.6	25.4	18.8
U3 - Toxicity prior Beta(0.5, 0.5)	19	11.6	17.6	11.4	21.7	18.7
U2 - Toxicity prior Beta(1, 1)	14.1	10.5	18.7	12.7	25.4	18.6
U3 - Toxicity prior Beta(1, 1)	17.8	12	17.1	11.9	22.3	18.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.2	4.6	5.5	3.8	4.5	
U3 - Toxicity prior Beta(0.15, 0.85)	9.2	4.7	5.4	3.9	4.4	
U2 - Toxicity prior Beta(0.3, 0.7)	9.1	4.6	5.2	3.9	4.8	
U3 - Toxicity prior Beta(0.3, 0.7)	9.1	4.6	5.3	3.9	4.8	
U2 - Toxicity prior Beta(0.5, 0.5)	8.7	4.2	5.4	4.2	5.2	
U3 - Toxicity prior Beta(0.5, 0.5)	8.6	4.3	5.5	4.3	5.2	
U2 - Toxicity prior Beta(1, 1)	8.7	4.2	5.5	4.2	5.3	
U3 - Toxicity prior Beta(1, 1)	8.7	4.2	5.4	4.2	5.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2.7	1.3	1.4	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2.7	1.3	1.4	0.6	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.7	1.3	1.3	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	2.7	1.3	1.4	0.6	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.5	1.5	1.4	0.7	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.5	1.5	1.4	0.7	
U2 - Toxicity prior Beta(1, 1)	0.4	2.5	1.4	1.4	0.7	
U3 - Toxicity prior Beta(1, 1)	0.4	2.5	1.5	1.4	0.7	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	12.6	31	11.1	3.7	1.5	40.1
U3 - Toxicity prior Beta(0.15, 0.85)	14	32.9	9	2.8	1.2	40.1
U2 - Toxicity prior Beta(0.3, 0.7)	12.1	30.5	12.4	4	1.4	39.6
U3 - Toxicity prior Beta(0.3, 0.7)	12.6	31.9	11.3	3.1	1.5	39.6
U2 - Toxicity prior Beta(0.5, 0.5)	10.9	29.2	14.9	4.8	1.8	38.4
U3 - Toxicity prior Beta(0.5, 0.5)	12.5	31.3	12.3	3.8	1.7	38.4
U2 - Toxicity prior Beta(1, 1)	11.3	28.3	15.1	4.6	1.9	38.8
U3 - Toxicity prior Beta(1, 1)	12.5	30.2	12.8	4.1	1.6	38.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	12.5	5.9	3.8	1.8	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	12.5	6	3.6	1.9	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	12.4	5.9	3.8	1.9	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	12.3	5.9	3.8	1.9	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	11.5	6.3	4.2	2.2	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	11.6	6.2	4.1	2.1	1.2	
U2 - Toxicity prior Beta(1, 1)	11.6	6.1	4.1	2.2	1.1	
U3 - Toxicity prior Beta(1, 1)	11.6	6.2	4.1	2.2	1.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.3	0.8	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.3	0.7	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.2	0.8	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.2	0.8	0.5	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.1	0.9	0.5	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.1	0.9	0.5	0.2	
U2 - Toxicity prior Beta(1, 1)	0.1	2.1	0.9	0.5	0.2	
U3 - Toxicity prior Beta(1, 1)	0.1	2.1	0.9	0.5	0.2	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	66.3	23.3	2.7	0.1	0	7.6
U3 - Toxicity prior Beta(0.15, 0.85)	69.5	21.7	1.5	0.2	0	7.1
U2 - Toxicity prior Beta(0.3, 0.7)	64.9	24.3	3.6	0.2	0	7
U3 - Toxicity prior Beta(0.3, 0.7)	68.1	22.8	1.7	0.3	0	7.1
U2 - Toxicity prior Beta(0.5, 0.5)	65.5	23.1	4.3	0.5	0	6.6
U3 - Toxicity prior Beta(0.5, 0.5)	69.8	20.1	3.1	0.3	0	6.7
U2 - Toxicity prior Beta(1, 1)	65.5	23.4	4.1	0.3	0	6.7
U3 - Toxicity prior Beta(1, 1)	69.1	21.1	2.7	0.2	0	6.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	19.2	5.9	3	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	19.2	5.9	2.9	0.7	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	19	5.9	3	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	19	5.9	3	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	18.2	6.3	3.3	1.1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	18.1	6.3	3.2	1.1	0.1	
U2 - Toxicity prior Beta(1, 1)	18.2	6.3	3.2	1.1	0.1	
U3 - Toxicity prior Beta(1, 1)	18.2	6.3	3.2	1.1	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	2.4	0.7	0.3	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	2.4	0.7	0.3	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	2.3	0.7	0.3	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	2.3	0.7	0.3	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	0.9	0.4	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2.2	0.9	0.4	0	
U2 - Toxicity prior Beta(1, 1)	0.7	2.2	0.8	0.4	0	
U3 - Toxicity prior Beta(1, 1)	0.7	2.2	0.8	0.4	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	15	14.4	11.3	36.6	19.5	3.2
U3 - Toxicity prior Beta(0.15, 0.85)	22	14.8	8.8	33.6	17.6	3.2
U2 - Toxicity prior Beta(0.3, 0.7)	15.4	12.4	10.8	37.5	20.7	3.2
U3 - Toxicity prior Beta(0.3, 0.7)	20	14.3	7.8	35.9	18.8	3.2
U2 - Toxicity prior Beta(0.5, 0.5)	12.4	11.5	9.5	39.4	23.9	3.3
U3 - Toxicity prior Beta(0.5, 0.5)	17.7	13.1	7.3	37.3	21.3	3.3
U2 - Toxicity prior Beta(1, 1)	11.6	11.3	8.7	40.2	24.9	3.3
U3 - Toxicity prior Beta(1, 1)	18.4	11.7	7.5	37.5	21.6	3.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.2	4.9	5.6	7.4	6.5	
U3 - Toxicity prior Beta(0.15, 0.85)	5.2	4.8	5.6	7.3	6.6	
U2 - Toxicity prior Beta(0.3, 0.7)	5.1	4.8	5.2	7.5	6.9	
U3 - Toxicity prior Beta(0.3, 0.7)	5	4.8	5.2	7.6	7	
U2 - Toxicity prior Beta(0.5, 0.5)	4.8	4.3	4.9	8.2	7.4	
U3 - Toxicity prior Beta(0.5, 0.5)	4.8	4.4	4.9	8.2	7.3	
U2 - Toxicity prior Beta(1, 1)	4.8	4.4	4.8	8.2	7.4	
U3 - Toxicity prior Beta(1, 1)	4.8	4.3	4.8	8.2	7.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3	1.4	2.3	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.9	1.4	2.2	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3	1.4	2.2	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	1.4	2.1	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.8	1.4	2.2	0.8	
U2 - Toxicity prior Beta(1, 1)	0.3	2.8	1.4	2.2	0.8	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	1.4	2.1	0.8	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	14.8	33.2	12	3.5	1.3	35.2
U3 - Toxicity prior Beta(0.15, 0.85)	16.4	33.3	10.8	3.2	1.1	35.2
U2 - Toxicity prior Beta(0.3, 0.7)	14.1	32.8	13	4.3	1.3	34.5
U3 - Toxicity prior Beta(0.3, 0.7)	15.2	34.5	10.9	3.8	1.2	34.4
U2 - Toxicity prior Beta(0.5, 0.5)	13.2	31.4	14.6	5.7	1.4	33.7
U3 - Toxicity prior Beta(0.5, 0.5)	14.9	32.3	13.3	4.5	1.3	33.7
U2 - Toxicity prior Beta(1, 1)	13.5	30.9	14.8	5.4	1.7	33.7
U3 - Toxicity prior Beta(1, 1)	15.1	32.6	12.4	4.9	1.2	33.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	12.3	6.3	4	2.1	0.8	
U3 - Toxicity prior Beta(0.15, 0.85)	12.3	6.2	4.1	2.1	0.8	
U2 - Toxicity prior Beta(0.3, 0.7)	12.1	6.1	4.1	2.2	1	
U3 - Toxicity prior Beta(0.3, 0.7)	12	6.2	4	2.2	1	
U2 - Toxicity prior Beta(0.5, 0.5)	11.2	6.6	4.5	2.4	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	11.2	6.6	4.4	2.5	1.1	
U2 - Toxicity prior Beta(1, 1)	11.2	6.5	4.5	2.4	1.1	
U3 - Toxicity prior Beta(1, 1)	11.2	6.5	4.5	2.4	1.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	0.8	0.6	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.6	0.8	0.6	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.5	0.9	0.6	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.5	0.9	0.6	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.3	1	0.7	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.4	1	0.7	0.2	
U2 - Toxicity prior Beta(1, 1)	0.2	2.4	1	0.7	0.2	
U3 - Toxicity prior Beta(1, 1)	0.2	2.3	1	0.7	0.2	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.2	17.6	13.7	26.9	12.1	7.5
U3 - Toxicity prior Beta(0.15, 0.85)	31.6	18.8	10.2	23.4	8.6	7.4
U2 - Toxicity prior Beta(0.3, 0.7)	20.7	16.4	12.8	28.7	14.1	7.3
U3 - Toxicity prior Beta(0.3, 0.7)	29.7	17.6	9.6	24.2	11.7	7.2
U2 - Toxicity prior Beta(0.5, 0.5)	18.2	13.8	14.7	30.8	15.3	7.2
U3 - Toxicity prior Beta(0.5, 0.5)	28	16.9	9.4	26.7	11.8	7.2
U2 - Toxicity prior Beta(1, 1)	18	13	14.9	30.4	16.5	7.2
U3 - Toxicity prior Beta(1, 1)	29.5	15.3	9.3	26.6	12.1	7.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	7.2	5.6	5.9	5.8	4.5	
U3 - Toxicity prior Beta(0.15, 0.85)	7.2	5.5	5.9	5.8	4.5	
U2 - Toxicity prior Beta(0.3, 0.7)	7	5.5	5.7	5.9	4.8	
U3 - Toxicity prior Beta(0.3, 0.7)	6.9	5.5	5.6	6.1	4.9	
U2 - Toxicity prior Beta(0.5, 0.5)	6.5	5	5.7	6.6	5.3	
U3 - Toxicity prior Beta(0.5, 0.5)	6.5	5.1	5.6	6.5	5.3	
U2 - Toxicity prior Beta(1, 1)	6.5	5	5.6	6.6	5.4	
U3 - Toxicity prior Beta(1, 1)	6.5	5.1	5.5	6.7	5.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	3.2	1.5	2	0.7	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	3.2	1.5	2	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	3.2	1.5	2	0.8	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.2	1.5	2	0.7	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.9	1.6	2	0.8	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.9	1.6	2	0.8	
U2 - Toxicity prior Beta(1, 1)	0.5	2.9	1.6	2	0.8	
U3 - Toxicity prior Beta(1, 1)	0.5	2.9	1.6	2	0.8	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	51.4	27.9	9.6	3.1	0.6	7.4
U3 - Toxicity prior Beta(0.15, 0.85)	60.7	25	5.3	1.6	0.2	7.2
U2 - Toxicity prior Beta(0.3, 0.7)	51.9	27.2	9.4	3.9	0.7	6.9
U3 - Toxicity prior Beta(0.3, 0.7)	62.4	23.7	4.7	2.1	0.2	6.9
U2 - Toxicity prior Beta(0.5, 0.5)	50.2	26.2	11.4	4.2	0.9	7.1
U3 - Toxicity prior Beta(0.5, 0.5)	63	21.4	5.8	2.1	0.5	7.2
U2 - Toxicity prior Beta(1, 1)	50.6	24.9	12.1	4.5	0.8	7.1
U3 - Toxicity prior Beta(1, 1)	62	22.5	5.9	2.1	0.3	7.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.9	7.3	5.6	3	1.2	
U3 - Toxicity prior Beta(0.15, 0.85)	11.8	7.4	5.7	3.1	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	11.7	7.2	5.7	3.1	1.3	
U3 - Toxicity prior Beta(0.3, 0.7)	11.7	7.2	5.7	3.2	1.3	
U2 - Toxicity prior Beta(0.5, 0.5)	11.3	6.9	5.7	3.7	1.5	
U3 - Toxicity prior Beta(0.5, 0.5)	11.3	6.8	5.8	3.7	1.6	
U2 - Toxicity prior Beta(1, 1)	11.2	6.9	5.8	3.7	1.6	
U3 - Toxicity prior Beta(1, 1)	11.2	6.8	5.8	3.7	1.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	3.5	1.5	1.2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	1	3.5	1.5	1.2	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	3.5	1.5	1.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	1.1	3.5	1.5	1.2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	3.2	1.7	1.3	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	1.1	3.2	1.7	1.4	0.4	
U2 - Toxicity prior Beta(1, 1)	1	3.2	1.7	1.3	0.4	
U3 - Toxicity prior Beta(1, 1)	1.1	3.2	1.7	1.3	0.4	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	31	25.4	16.3	16.7	10.5	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	38.4	29.7	14	12.2	5.6	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	29	27.4	16.6	15.5	11.4	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	37.3	29.6	14.4	13.4	5.2	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	28.7	26.6	16.8	16.3	11.5	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	35.4	30.1	14.3	13.3	6.8	0.1
U2 - Toxicity prior Beta(1, 1)	30.1	26	16.1	15.5	12.2	0.1
U3 - Toxicity prior Beta(1, 1)	36.9	29.8	14	13.1	6.1	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.8	5.7	5.5	5.9	7	
U3 - Toxicity prior Beta(0.15, 0.85)	5.9	5.7	5.4	5.9	7	
U2 - Toxicity prior Beta(0.3, 0.7)	5.8	5.6	5.4	5.8	7.2	
U3 - Toxicity prior Beta(0.3, 0.7)	5.8	5.8	5.4	5.8	7.2	
U2 - Toxicity prior Beta(0.5, 0.5)	5.7	5.7	5.4	5.8	7.3	
U3 - Toxicity prior Beta(0.5, 0.5)	5.8	5.7	5.4	5.8	7.3	
U2 - Toxicity prior Beta(1, 1)	5.8	5.6	5.5	5.7	7.3	
U3 - Toxicity prior Beta(1, 1)	5.8	5.6	5.4	5.8	7.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	1	3.2	1.8	2.7	1.5	
U3 - Toxicity prior Beta(0.15, 0.85)	1	3.2	1.8	2.7	1.5	
U2 - Toxicity prior Beta(0.3, 0.7)	1	3.2	1.8	2.7	1.5	
U3 - Toxicity prior Beta(0.3, 0.7)	1	3.2	1.8	2.7	1.5	
U2 - Toxicity prior Beta(0.5, 0.5)	1	3.2	1.8	2.6	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	1	3.2	1.8	2.6	1.4	
U2 - Toxicity prior Beta(1, 1)	1	3.2	1.9	2.6	1.5	
U3 - Toxicity prior Beta(1, 1)	1	3.2	1.8	2.6	1.5	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	29.2	53.2	16.1	1.2	0	0.3
U3 - Toxicity prior Beta(0.15, 0.85)	31.3	54.1	13.3	1	0	0.3
U2 - Toxicity prior Beta(0.3, 0.7)	29.3	52.4	16.6	1.4	0	0.3
U3 - Toxicity prior Beta(0.3, 0.7)	32.8	51.3	14.9	0.7	0	0.3
U2 - Toxicity prior Beta(0.5, 0.5)	27.8	51.6	19.2	1.1	0	0.3
U3 - Toxicity prior Beta(0.5, 0.5)	31.6	52.2	15	0.9	0	0.3
U2 - Toxicity prior Beta(1, 1)	28.9	51.1	18.7	1	0	0.3
U3 - Toxicity prior Beta(1, 1)	31	52	15.9	0.8	0	0.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.6	10.3	6.8	2	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	10.5	10.3	6.9	1.9	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	10.3	10	7	2.2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	10.4	10	7	2.2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	9.7	10	7.2	2.7	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	9.7	10	7.2	2.7	0.4	
U2 - Toxicity prior Beta(1, 1)	9.7	10	7.1	2.7	0.4	
U3 - Toxicity prior Beta(1, 1)	9.7	9.9	7.2	2.7	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	2.9	1.1	0.5	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	2.9	1.1	0.5	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.9	1.1	0.5	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.9	1.1	0.5	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.8	1.2	0.6	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.8	1.2	0.6	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	2.8	1.2	0.6	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	2.8	1.2	0.6	0.1	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	8.5	10	11.1	16.4	53.3	0.7
U3 - Toxicity prior Beta(0.15, 0.85)	15.3	11.7	11.2	14.5	46.6	0.7
U2 - Toxicity prior Beta(0.3, 0.7)	9.4	10	9.8	16.2	53.9	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	14.9	12.2	10.2	13.7	48.3	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	8.4	9.1	9.7	14.9	57.2	0.7
U3 - Toxicity prior Beta(0.5, 0.5)	13.4	11.4	9.4	14	51.1	0.7
U2 - Toxicity prior Beta(1, 1)	8.5	8.3	8.5	14.2	59.8	0.7
U3 - Toxicity prior Beta(1, 1)	12.8	11.2	9.5	13.8	52	0.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.8	4	4.8	5.8	11.6	
U3 - Toxicity prior Beta(0.15, 0.85)	3.8	4	4.8	5.6	11.7	
U2 - Toxicity prior Beta(0.3, 0.7)	3.8	3.9	4.6	5.7	11.9	
U3 - Toxicity prior Beta(0.3, 0.7)	3.8	3.9	4.6	5.6	11.9	
U2 - Toxicity prior Beta(0.5, 0.5)	3.7	3.9	4.5	5.7	12.2	
U3 - Toxicity prior Beta(0.5, 0.5)	3.7	3.8	4.5	5.6	12.3	
U2 - Toxicity prior Beta(1, 1)	3.8	3.9	4.5	5.6	12.2	
U3 - Toxicity prior Beta(1, 1)	3.7	3.9	4.5	5.7	12.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.4	2.4	1.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.5	2.4	1.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.3	1.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.8	1.4	2.3	1.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.2	2.8	1.4	2.4	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	0.2	2.7	1.4	2.3	1.4	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	2.3	1.4	
U3 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	2.3	1.4	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	19.7	54.7	15.8	1.1	0.1	8.6
U3 - Toxicity prior Beta(0.15, 0.85)	22.4	55.2	13	0.7	0	8.7
U2 - Toxicity prior Beta(0.3, 0.7)	19.4	54.4	16	1.8	0.1	8.3
U3 - Toxicity prior Beta(0.3, 0.7)	21.9	53.9	14.3	1.6	0	8.3
U2 - Toxicity prior Beta(0.5, 0.5)	18.7	52.4	18.5	2.1	0.1	8.2
U3 - Toxicity prior Beta(0.5, 0.5)	20.7	52.4	17.2	1.5	0.1	8.1
U2 - Toxicity prior Beta(1, 1)	18.3	52.5	19.6	1.4	0.1	8.1
U3 - Toxicity prior Beta(1, 1)	21.1	52.6	16.9	1.3	0	8.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.6	10.6	6.7	1.7	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	9.7	10.6	6.6	1.7	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	9.5	10.3	6.9	2	0.3	
U3 - Toxicity prior Beta(0.3, 0.7)	9.5	10.2	6.8	2	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	8.8	10.3	7	2.6	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	8.8	10.2	7.1	2.6	0.3	
U2 - Toxicity prior Beta(1, 1)	8.8	10.3	7.1	2.6	0.4	
U3 - Toxicity prior Beta(1, 1)	8.8	10.2	7.1	2.6	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.6	1.2	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.7	1.2	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.6	1.2	0.6	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.6	1.3	0.6	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.4	1.4	0.7	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.3	1.4	0.7	0.1	
U2 - Toxicity prior Beta(1, 1)	0.4	3.3	1.4	0.7	0.1	
U3 - Toxicity prior Beta(1, 1)	0.4	3.3	1.4	0.7	0.1	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.3	48.2	27	8.3	0.9	4.3
U3 - Toxicity prior Beta(0.15, 0.85)	16.8	49	22.7	6.7	0.5	4.3
U2 - Toxicity prior Beta(0.3, 0.7)	11.5	46.6	27.4	9.4	0.8	4.3
U3 - Toxicity prior Beta(0.3, 0.7)	15.7	48	23.8	7.7	0.5	4.3
U2 - Toxicity prior Beta(0.5, 0.5)	11.9	44.2	26.8	11.3	1.6	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	16.4	45.8	23.5	9.4	0.7	4.2
U2 - Toxicity prior Beta(1, 1)	11.7	44.8	26.1	12	1.2	4.2
U3 - Toxicity prior Beta(1, 1)	17.7	44.8	22.8	9.4	1	4.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	6	9.1	8.4	4.2	1.7	
U3 - Toxicity prior Beta(0.15, 0.85)	6	9.1	8.4	4.1	1.8	
U2 - Toxicity prior Beta(0.3, 0.7)	5.8	8.8	8.3	4.5	1.9	
U3 - Toxicity prior Beta(0.3, 0.7)	5.8	8.7	8.4	4.6	1.9	
U2 - Toxicity prior Beta(0.5, 0.5)	5.5	8.4	8	5.4	2.2	
U3 - Toxicity prior Beta(0.5, 0.5)	5.5	8.3	8.1	5.3	2.2	
U2 - Toxicity prior Beta(1, 1)	5.5	8.3	8.1	5.3	2.1	
U3 - Toxicity prior Beta(1, 1)	5.5	8.3	8.1	5.4	2.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.7	1.8	1.5	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.7	1.8	1.5	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.7	1.8	1.5	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3.6	1.7	1.5	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	3.5	1.9	1.6	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	3.5	1.9	1.6	0.5	
U2 - Toxicity prior Beta(1, 1)	0.4	3.5	1.9	1.6	0.4	
U3 - Toxicity prior Beta(1, 1)	0.4	3.5	1.9	1.6	0.5	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	58.9	1.3	0	0	0	39.8
U3 - Toxicity prior Beta(0.15, 0.85)	58.4	1.3	0	0	0	40.3
U2 - Toxicity prior Beta(0.3, 0.7)	58.7	1.4	0	0	0	39.9
U3 - Toxicity prior Beta(0.3, 0.7)	59	1.3	0	0	0	39.7
U2 - Toxicity prior Beta(0.5, 0.5)	59.7	1.2	0	0	0	39.1
U3 - Toxicity prior Beta(0.5, 0.5)	59.7	1.1	0	0	0	39.2
U2 - Toxicity prior Beta(1, 1)	59.8	1.3	0	0	0	38.9
U3 - Toxicity prior Beta(1, 1)	59.8	1.2	0	0	0	39
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	21	1.9	0.4	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	21	1.9	0.4	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	21.1	1.9	0.5	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	21.1	1.9	0.4	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	20.9	2.1	0.5	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	20.9	2.1	0.5	0	0	
U2 - Toxicity prior Beta(1, 1)	20.9	2.1	0.5	0	0	
U3 - Toxicity prior Beta(1, 1)	20.9	2.1	0.5	0	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	0.7	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	0.7	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	0.7	0.1	0	0	
U2 - Toxicity prior Beta(1, 1)	0.1	0.7	0.1	0	0	
U3 - Toxicity prior Beta(1, 1)	0.1	0.7	0.1	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	24.3	59.5	10.4	1.1	0	4.7
U3 - Toxicity prior Beta(0.15, 0.85)	29.1	56.4	8.6	1.1	0	4.8
U2 - Toxicity prior Beta(0.3, 0.7)	24.6	59	10.2	1.7	0	4.5
U3 - Toxicity prior Beta(0.3, 0.7)	28.6	56.6	8.9	1.4	0	4.5
U2 - Toxicity prior Beta(0.5, 0.5)	23.3	57.2	12.9	2.4	0	4.2
U3 - Toxicity prior Beta(0.5, 0.5)	26.6	55.8	11.2	2	0	4.4
U2 - Toxicity prior Beta(1, 1)	23.2	56.8	13.4	2.2	0	4.4
U3 - Toxicity prior Beta(1, 1)	27.1	55.3	11.4	2	0	4.2
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.2	12.7	5.9	1.3	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	9.2	12.6	6	1.3	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	9	12.5	6.1	1.5	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	9.1	12.3	6.1	1.6	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	8.3	12.6	6.1	2.2	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	8.4	12.4	6.1	2.2	0.3	
U2 - Toxicity prior Beta(1, 1)	8.4	12.3	6.2	2.3	0.3	
U3 - Toxicity prior Beta(1, 1)	8.4	12.3	6.1	2.3	0.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	4	1.2	0.4	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	4	1.2	0.5	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	4	1.2	0.4	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	4	1.3	0.4	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.6	3.7	1.4	0.5	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.6	1.4	0.5	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3.6	1.4	0.5	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	3.6	1.4	0.5	0.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	18.1	6.7	0	0	0	75.2
U3 - Toxicity prior Beta(0.15, 0.85)	19	5.7	0	0	0	75.3
U2 - Toxicity prior Beta(0.3, 0.7)	18	6.5	0	0	0	75.5
U3 - Toxicity prior Beta(0.3, 0.7)	19.2	5.5	0	0	0	75.3
U2 - Toxicity prior Beta(0.5, 0.5)	19.5	4.6	0	0	0	75.9
U3 - Toxicity prior Beta(0.5, 0.5)	19.5	3.8	0	0	0	76.7
U2 - Toxicity prior Beta(1, 1)	19	4.7	0	0	0	76.3
U3 - Toxicity prior Beta(1, 1)	19.7	3.8	0	0	0	76.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	17.3	2.3	0.4	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	17.4	2.3	0.4	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	17.4	2.3	0.4	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	17.3	2.4	0.4	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	17.2	2.7	0.5	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	17.2	2.7	0.5	0	0	
U2 - Toxicity prior Beta(1, 1)	17.2	2.7	0.5	0	0	
U3 - Toxicity prior Beta(1, 1)	17.2	2.7	0.5	0	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	1.4	0.1	0	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	1.4	0.1	0	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	1.4	0.1	0	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	1.4	0.1	0	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	1.4	0.2	0	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	1.4	0.2	0	0	
U2 - Toxicity prior Beta(1, 1)	0.4	1.4	0.2	0	0	
U3 - Toxicity prior Beta(1, 1)	0.4	1.4	0.2	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	1.1	1.9	6	48.1	42.8	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	2.3	2.6	7.6	49.4	38	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	1.1	1.9	5.5	46.6	44.8	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	2.2	2.6	8	49	38.1	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	1.1	1.3	5.7	46.3	45.6	0
U3 - Toxicity prior Beta(0.5, 0.5)	2.2	2.2	7.4	49	39.2	0
U2 - Toxicity prior Beta(1, 1)	1	1.7	5.4	47.4	44.5	0
U3 - Toxicity prior Beta(1, 1)	2.7	2.7	7.5	48	39.1	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.3	3.2	3.8	8.7	11.1	
U3 - Toxicity prior Beta(0.15, 0.85)	3.2	3.2	3.8	8.7	11.2	
U2 - Toxicity prior Beta(0.3, 0.7)	3.2	3.2	3.8	8.5	11.3	
U3 - Toxicity prior Beta(0.3, 0.7)	3.2	3.2	3.8	8.6	11.2	
U2 - Toxicity prior Beta(0.5, 0.5)	3.2	3.1	3.7	8.6	11.4	
U3 - Toxicity prior Beta(0.5, 0.5)	3.2	3.2	3.7	8.5	11.3	
U2 - Toxicity prior Beta(1, 1)	3.2	3.2	3.7	8.6	11.3	
U3 - Toxicity prior Beta(1, 1)	3.2	3.2	3.7	8.6	11.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.2	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.1	2.5	1.2	3.2	1.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.2	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.5	1.2	3.2	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.2	1.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.5	1.2	3.2	1.2	
U2 - Toxicity prior Beta(1, 1)	0.1	2.5	1.2	3.2	1.2	
U3 - Toxicity prior Beta(1, 1)	0.1	2.5	1.2	3.2	1.2	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	34.7	31	28.8	4.8	0	0.7
U3 - Toxicity prior Beta(0.15, 0.85)	38.3	31.9	24.7	4.4	0	0.7
U2 - Toxicity prior Beta(0.3, 0.7)	34.7	30.8	29.3	4.5	0	0.7
U3 - Toxicity prior Beta(0.3, 0.7)	38.9	31.2	25	4.2	0	0.7
U2 - Toxicity prior Beta(0.5, 0.5)	33.5	28.2	32.7	5	0	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	39.1	28.3	27.2	4.8	0	0.6
U2 - Toxicity prior Beta(1, 1)	33	27.9	33.5	5	0	0.6
U3 - Toxicity prior Beta(1, 1)	39.1	27.1	28.9	4.3	0	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.7	8.5	8.9	2.4	0.4	
U3 - Toxicity prior Beta(0.15, 0.85)	9.6	8.7	8.9	2.4	0.4	
U2 - Toxicity prior Beta(0.3, 0.7)	9.4	8.2	9.1	2.8	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	9.4	8.1	9.2	2.8	0.4	
U2 - Toxicity prior Beta(0.5, 0.5)	9	7.6	9.4	3.4	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	9	7.7	9.2	3.4	0.5	
U2 - Toxicity prior Beta(1, 1)	9	7.7	9.3	3.4	0.5	
U3 - Toxicity prior Beta(1, 1)	9	7.7	9.2	3.4	0.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	3.1	1.5	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	3.2	1.5	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	3.1	1.5	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	3.1	1.5	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2.9	1.6	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2.9	1.6	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	0.7	3	1.6	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	0.7	2.9	1.6	0.8	0.1	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	30.7	30.1	24.3	4.3	0	10.6
U3 - Toxicity prior Beta(0.15, 0.85)	33.4	30.4	21.7	4.1	0	10.4
U2 - Toxicity prior Beta(0.3, 0.7)	29.2	29	26	5.5	0	10.3
U3 - Toxicity prior Beta(0.3, 0.7)	33.6	29.9	21.7	4.5	0	10.3
U2 - Toxicity prior Beta(0.5, 0.5)	29.3	25.9	28	6.7	0	10.1
U3 - Toxicity prior Beta(0.5, 0.5)	32.6	26.5	24.7	6.2	0	10
U2 - Toxicity prior Beta(1, 1)	28.4	26.5	28.6	6.4	0	10.1
U3 - Toxicity prior Beta(1, 1)	32.5	25.9	25.3	6.2	0	10.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	11.6	7.1	7.5	2.3	0.3	
U3 - Toxicity prior Beta(0.15, 0.85)	11.6	7.1	7.5	2.3	0.3	
U2 - Toxicity prior Beta(0.3, 0.7)	11.3	6.8	7.7	2.6	0.4	
U3 - Toxicity prior Beta(0.3, 0.7)	11.4	6.9	7.6	2.6	0.3	
U2 - Toxicity prior Beta(0.5, 0.5)	10.7	6.7	8	3.2	0.4	
U3 - Toxicity prior Beta(0.5, 0.5)	10.6	6.6	8.1	3.2	0.4	
U2 - Toxicity prior Beta(1, 1)	10.7	6.6	8.1	3.2	0.4	
U3 - Toxicity prior Beta(1, 1)	10.7	6.6	8.1	3.2	0.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.1	1.2	0.8	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.1	1.2	0.8	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	2.9	1.2	0.8	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	3	1.2	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	2.7	1.4	0.8	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	2.7	1.4	0.8	0.1	
U2 - Toxicity prior Beta(1, 1)	0.5	2.7	1.4	0.8	0.1	
U3 - Toxicity prior Beta(1, 1)	0.5	2.7	1.4	0.8	0.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	21.2	25.8	39.7	7.1	0.7	5.5
U3 - Toxicity prior Beta(0.15, 0.85)	25.1	25.9	36.7	6.6	0.2	5.5
U2 - Toxicity prior Beta(0.3, 0.7)	20.9	23.3	41.1	8.7	0.7	5.3
U3 - Toxicity prior Beta(0.3, 0.7)	25	24.8	36.8	7.7	0.4	5.3
U2 - Toxicity prior Beta(0.5, 0.5)	20.7	20.9	42.4	9.8	0.9	5.3
U3 - Toxicity prior Beta(0.5, 0.5)	24.9	23.3	37.1	8.8	0.6	5.3
U2 - Toxicity prior Beta(1, 1)	20.1	21.1	42.8	9.8	0.9	5.3
U3 - Toxicity prior Beta(1, 1)	24.9	22.4	38.2	8.7	0.5	5.3
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	8.5	6.9	9.2	3.6	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	8.5	6.8	9.3	3.6	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	8.4	6.3	9.3	4	1.2	
U3 - Toxicity prior Beta(0.3, 0.7)	8.4	6.5	9.2	4	1.2	
U2 - Toxicity prior Beta(0.5, 0.5)	7.9	5.9	9.4	4.8	1.4	
U3 - Toxicity prior Beta(0.5, 0.5)	8	6	9.2	4.9	1.4	
U2 - Toxicity prior Beta(1, 1)	7.9	5.9	9.3	4.9	1.4	
U3 - Toxicity prior Beta(1, 1)	7.9	5.9	9.3	4.9	1.4	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3	1.6	1	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3	1.6	1	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.9	1.6	1	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	3	1.5	1	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.8	1.7	1.1	0.3	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.8	1.7	1.1	0.3	
U2 - Toxicity prior Beta(1, 1)	0.4	2.8	1.7	1.1	0.3	
U3 - Toxicity prior Beta(1, 1)	0.4	2.8	1.7	1.1	0.3	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	22.5	38.1	4.3	0	0	35.1
U3 - Toxicity prior Beta(0.15, 0.85)	23.2	38.2	3.9	0	0	34.7
U2 - Toxicity prior Beta(0.3, 0.7)	21.7	38.9	4.9	0.1	0	34.4
U3 - Toxicity prior Beta(0.3, 0.7)	22.6	38.9	4.4	0	0	34.1
U2 - Toxicity prior Beta(0.5, 0.5)	21.2	39.2	6.1	0.3	0	33.2
U3 - Toxicity prior Beta(0.5, 0.5)	22.9	38.7	4.9	0.4	0	33.1
U2 - Toxicity prior Beta(1, 1)	21.6	39.3	5.4	0.4	0	33.3
U3 - Toxicity prior Beta(1, 1)	22.9	38.8	4.8	0.4	0	33.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	12.9	8.3	3.5	0.6	0.1	
U3 - Toxicity prior Beta(0.15, 0.85)	12.9	8.4	3.5	0.6	0.1	
U2 - Toxicity prior Beta(0.3, 0.7)	12.6	8.4	3.7	0.7	0.1	
U3 - Toxicity prior Beta(0.3, 0.7)	12.7	8.3	3.8	0.8	0.1	
U2 - Toxicity prior Beta(0.5, 0.5)	11.8	9	4	1.1	0.1	
U3 - Toxicity prior Beta(0.5, 0.5)	11.9	8.9	4	1	0.1	
U2 - Toxicity prior Beta(1, 1)	11.8	9	3.9	1.1	0.1	
U3 - Toxicity prior Beta(1, 1)	11.9	8.9	3.9	1.1	0.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.7	0.7	0.2	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.7	0.7	0.2	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.7	0.8	0.2	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.7	0.8	0.2	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.5	0.9	0.3	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.5	0.9	0.3	0	
U2 - Toxicity prior Beta(1, 1)	0.3	2.5	0.9	0.3	0	
U3 - Toxicity prior Beta(1, 1)	0.3	2.5	0.9	0.3	0	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	68.5	25.2	3.2	0.1	0	3
U3 - Toxicity prior Beta(0.15, 0.85)	71.7	22.6	2.8	0.2	0	2.7
U2 - Toxicity prior Beta(0.3, 0.7)	67.7	25.9	3.5	0.3	0	2.6
U3 - Toxicity prior Beta(0.3, 0.7)	71.1	23.1	2.9	0.2	0	2.7
U2 - Toxicity prior Beta(0.5, 0.5)	66.6	27.1	3.8	0.2	0	2.3
U3 - Toxicity prior Beta(0.5, 0.5)	69.6	25.2	2.5	0.2	0	2.5
U2 - Toxicity prior Beta(1, 1)	66.6	27.1	3.7	0.2	0	2.4
U3 - Toxicity prior Beta(1, 1)	69.9	25	2.6	0.1	0	2.4
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	18.4	7.5	3.1	0.4	0	
U3 - Toxicity prior Beta(0.15, 0.85)	18.4	7.5	3.1	0.4	0	
U2 - Toxicity prior Beta(0.3, 0.7)	18.4	7.5	3.2	0.5	0	
U3 - Toxicity prior Beta(0.3, 0.7)	18.3	7.5	3.2	0.5	0	
U2 - Toxicity prior Beta(0.5, 0.5)	17.7	7.9	3.4	0.6	0	
U3 - Toxicity prior Beta(0.5, 0.5)	17.7	7.9	3.3	0.7	0	
U2 - Toxicity prior Beta(1, 1)	17.7	7.8	3.3	0.7	0	
U3 - Toxicity prior Beta(1, 1)	17.6	7.9	3.3	0.7	0	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.7	2.1	0.7	0.1	0	
U3 - Toxicity prior Beta(0.15, 0.85)	0.7	2.1	0.7	0.1	0	
U2 - Toxicity prior Beta(0.3, 0.7)	0.7	2.1	0.7	0.1	0	
U3 - Toxicity prior Beta(0.3, 0.7)	0.7	2.1	0.7	0.1	0	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	2	0.8	0.2	0	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	2	0.8	0.2	0	
U2 - Toxicity prior Beta(1, 1)	0.7	2	0.8	0.1	0	
U3 - Toxicity prior Beta(1, 1)	0.7	2	0.8	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20.3	16.2	14.9	36.4	11.4	0.8
U3 - Toxicity prior Beta(0.15, 0.85)	25.8	17.8	13.9	33.3	8.4	0.8
U2 - Toxicity prior Beta(0.3, 0.7)	18.9	15.8	14.9	38.2	11.4	0.8
U3 - Toxicity prior Beta(0.3, 0.7)	24.5	17.7	13.4	34.3	9.3	0.8
U2 - Toxicity prior Beta(0.5, 0.5)	17	13.9	12.6	42.3	13.4	0.8
U3 - Toxicity prior Beta(0.5, 0.5)	23.3	16.2	12.5	37.1	10.1	0.8
U2 - Toxicity prior Beta(1, 1)	17	13.9	12.9	41.4	14	0.8
U3 - Toxicity prior Beta(1, 1)	22.9	16.7	12.3	36.6	10.7	0.8
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.4	4.8	6.1	8.1	5.6	
U3 - Toxicity prior Beta(0.15, 0.85)	5.4	4.9	6	8	5.6	
U2 - Toxicity prior Beta(0.3, 0.7)	5.2	4.6	5.8	8.4	5.9	
U3 - Toxicity prior Beta(0.3, 0.7)	5.3	4.7	5.7	8.4	5.9	
U2 - Toxicity prior Beta(0.5, 0.5)	5	4.4	5.5	8.9	6.1	
U3 - Toxicity prior Beta(0.5, 0.5)	5.1	4.3	5.5	8.9	6.2	
U2 - Toxicity prior Beta(1, 1)	5	4.4	5.5	8.8	6.2	
U3 - Toxicity prior Beta(1, 1)	5	4.4	5.5	8.8	6.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	2.9	1.6	2.3	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	2.9	1.6	2.3	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	2.8	1.6	2.3	0.9	
U3 - Toxicity prior Beta(0.3, 0.7)	0.4	2.8	1.5	2.3	0.9	
U2 - Toxicity prior Beta(0.5, 0.5)	0.4	2.7	1.6	2.3	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	0.4	2.7	1.5	2.2	0.9	
U2 - Toxicity prior Beta(1, 1)	0.4	2.7	1.6	2.2	0.9	
U3 - Toxicity prior Beta(1, 1)	0.4	2.7	1.6	2.2	0.9	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	7	34.7	28	24.1	6.1	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	9.5	39.3	26.4	20.3	4.4	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	6.3	34.6	28.6	24.6	5.8	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	9.6	39.8	26.1	20.4	4	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	6.5	33.3	27.9	24.8	7.4	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	8.9	40.6	24.9	21.1	4.4	0.1
U2 - Toxicity prior Beta(1, 1)	6.4	32.7	28.5	24.8	7.5	0.1
U3 - Toxicity prior Beta(1, 1)	9.5	41	24.4	20.5	4.5	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.1	5.9	6.5	7.8	5.7	
U3 - Toxicity prior Beta(0.15, 0.85)	4.1	5.8	6.5	7.9	5.7	
U2 - Toxicity prior Beta(0.3, 0.7)	4.1	5.7	6.4	7.9	5.9	
U3 - Toxicity prior Beta(0.3, 0.7)	4.1	5.7	6.4	7.9	5.9	
U2 - Toxicity prior Beta(0.5, 0.5)	4	5.7	6.3	7.9	6	
U3 - Toxicity prior Beta(0.5, 0.5)	4.1	5.7	6.3	7.9	6	
U2 - Toxicity prior Beta(1, 1)	4.1	5.7	6.2	7.9	6.1	
U3 - Toxicity prior Beta(1, 1)	4	5.7	6.2	7.9	6.1	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.1	2.8	1.4	1.8	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.8	1.4	1.8	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	1.4	1.8	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.1	2.8	1.4	1.8	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.7	1.4	1.8	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.8	1.4	1.8	1.1	
U2 - Toxicity prior Beta(1, 1)	0.2	2.7	1.4	1.8	1.1	
U3 - Toxicity prior Beta(1, 1)	0.1	2.8	1.4	1.8	1.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	8.9	14.9	35.8	28.6	11.2	0.6
U3 - Toxicity prior Beta(0.15, 0.85)	13.7	19	35	25.3	6.4	0.6
U2 - Toxicity prior Beta(0.3, 0.7)	9.3	15.5	34.8	28.7	11.1	0.6
U3 - Toxicity prior Beta(0.3, 0.7)	15.1	17.5	36.1	24.3	6.4	0.6
U2 - Toxicity prior Beta(0.5, 0.5)	8.5	14.3	35.5	27.7	13.4	0.6
U3 - Toxicity prior Beta(0.5, 0.5)	14.6	19.1	34.5	24.3	6.9	0.6
U2 - Toxicity prior Beta(1, 1)	8	15.4	35.4	27.8	12.8	0.6
U3 - Toxicity prior Beta(1, 1)	13.9	18.8	34.2	25.1	7.4	0.6
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4	4.3	7.7	7.3	6.6	
U3 - Toxicity prior Beta(0.15, 0.85)	4	4.4	7.7	7.3	6.6	
U2 - Toxicity prior Beta(0.3, 0.7)	4	4.4	7.5	7.3	6.7	
U3 - Toxicity prior Beta(0.3, 0.7)	4	4.3	7.5	7.3	6.8	
U2 - Toxicity prior Beta(0.5, 0.5)	4	4.3	7.4	7.4	6.9	
U3 - Toxicity prior Beta(0.5, 0.5)	3.9	4.3	7.4	7.4	6.8	
U2 - Toxicity prior Beta(1, 1)	3.9	4.3	7.4	7.5	6.8	
U3 - Toxicity prior Beta(1, 1)	4	4.3	7.4	7.4	6.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	2	2.7	1.1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	2.9	2	2.6	1.1	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	2	2.6	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	2.9	2	2.6	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	2	2.6	1.1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	2.9	2.1	2.6	1.1	
U2 - Toxicity prior Beta(1, 1)	0.3	2.9	2	2.6	1.1	
U3 - Toxicity prior Beta(1, 1)	0.3	2.8	2	2.6	1.1	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	7.2	6.5	13.4	42.7	30.1	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	11.2	8.8	14	41.2	24.7	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	7.9	6.4	13.2	42.1	30.4	0
U3 - Toxicity prior Beta(0.3, 0.7)	11.1	8.2	14.1	41.1	25.5	0
U2 - Toxicity prior Beta(0.5, 0.5)	6.3	6	12.3	41.7	33.7	0
U3 - Toxicity prior Beta(0.5, 0.5)	10.6	7.5	12.1	41.8	28	0
U2 - Toxicity prior Beta(1, 1)	7.4	5.6	11.4	42.6	33	0
U3 - Toxicity prior Beta(1, 1)	10.5	7.8	12	42.1	27.6	0
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	3.5	3.7	5.3	8.4	9.2	
U3 - Toxicity prior Beta(0.15, 0.85)	3.5	3.6	5.3	8.4	9.2	
U2 - Toxicity prior Beta(0.3, 0.7)	3.5	3.7	5.1	8.2	9.5	
U3 - Toxicity prior Beta(0.3, 0.7)	3.5	3.6	5.1	8.3	9.5	
U2 - Toxicity prior Beta(0.5, 0.5)	3.4	3.5	4.8	8.6	9.6	
U3 - Toxicity prior Beta(0.5, 0.5)	3.4	3.5	4.9	8.6	9.6	
U2 - Toxicity prior Beta(1, 1)	3.4	3.6	4.8	8.6	9.6	
U3 - Toxicity prior Beta(1, 1)	3.4	3.6	4.8	8.6	9.6	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.9	1	
U3 - Toxicity prior Beta(0.15, 0.85)	0.2	2.7	1.4	2.9	0.9	
U2 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.8	1	
U3 - Toxicity prior Beta(0.3, 0.7)	0.2	2.7	1.4	2.8	1	
U2 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.8	1	
U3 - Toxicity prior Beta(0.5, 0.5)	0.1	2.6	1.4	2.8	1	
U2 - Toxicity prior Beta(1, 1)	0.1	2.6	1.4	2.8	1	
U3 - Toxicity prior Beta(1, 1)	0.2	2.6	1.4	2.8	1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	15	14.2	48	16.3	4.5	2
U3 - Toxicity prior Beta(0.15, 0.85)	21.3	15	43.8	14.9	2.9	2.1
U2 - Toxicity prior Beta(0.3, 0.7)	14	14.8	45.8	18.7	4.7	2
U3 - Toxicity prior Beta(0.3, 0.7)	21.5	15.8	41.8	16.3	2.6	2
U2 - Toxicity prior Beta(0.5, 0.5)	13.5	13.6	43.1	22.5	5.4	1.9
U3 - Toxicity prior Beta(0.5, 0.5)	18.9	16	39.8	20.1	3.3	1.9
U2 - Toxicity prior Beta(1, 1)	11.5	13.4	44	23.5	5.7	1.9
U3 - Toxicity prior Beta(1, 1)	18.3	16.4	40.1	19.6	3.7	1.9
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	5.5	6.1	9.8	5.4	2.9	
U3 - Toxicity prior Beta(0.15, 0.85)	5.5	6.1	9.9	5.4	2.9	
U2 - Toxicity prior Beta(0.3, 0.7)	5.4	5.8	9.8	5.5	3.2	
U3 - Toxicity prior Beta(0.3, 0.7)	5.3	5.8	9.9	5.6	3.2	
U2 - Toxicity prior Beta(0.5, 0.5)	5.1	5.5	9.3	6.3	3.6	
U3 - Toxicity prior Beta(0.5, 0.5)	5	5.5	9.3	6.3	3.6	
U2 - Toxicity prior Beta(1, 1)	5.1	5.3	9.3	6.5	3.6	
U3 - Toxicity prior Beta(1, 1)	5	5.5	9.3	6.4	3.5	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.5	3.3	1.9	1.6	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.5	3.3	2	1.6	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.5	3.2	1.9	1.6	0.6	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.2	1.9	1.6	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.1	2	1.7	0.6	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3	2.1	1.7	0.6	
U2 - Toxicity prior Beta(1, 1)	0.5	3	2.1	1.8	0.6	
U3 - Toxicity prior Beta(1, 1)	0.4	3	2.1	1.7	0.6	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	11.8	40.7	33.8	11	2.6	0.1
U3 - Toxicity prior Beta(0.15, 0.85)	15.7	48.7	25.3	8.6	1.6	0.1
U2 - Toxicity prior Beta(0.3, 0.7)	11.3	43.4	31.1	11.5	2.6	0.1
U3 - Toxicity prior Beta(0.3, 0.7)	14.7	47.6	26.8	9.3	1.5	0.1
U2 - Toxicity prior Beta(0.5, 0.5)	10	42	30.4	14.3	3.2	0.1
U3 - Toxicity prior Beta(0.5, 0.5)	14	48	24.7	11.4	1.8	0.1
U2 - Toxicity prior Beta(1, 1)	10.3	42.3	29.7	14	3.6	0.1
U3 - Toxicity prior Beta(1, 1)	14.8	47.8	24.3	11.1	1.9	0.1
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	4.6	8.7	8.9	5.1	2.7	
U3 - Toxicity prior Beta(0.15, 0.85)	4.6	8.8	8.8	5.1	2.7	
U2 - Toxicity prior Beta(0.3, 0.7)	4.6	8.4	8.7	5.3	3	
U3 - Toxicity prior Beta(0.3, 0.7)	4.6	8.5	8.7	5.2	3	
U2 - Toxicity prior Beta(0.5, 0.5)	4.5	7.9	8.4	6	3.2	
U3 - Toxicity prior Beta(0.5, 0.5)	4.4	8.1	8.3	6	3.2	
U2 - Toxicity prior Beta(1, 1)	4.5	7.9	8.3	6.1	3.2	
U3 - Toxicity prior Beta(1, 1)	4.4	8	8.3	6.1	3.2	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.3	3.1	1.6	1.4	0.5	
U3 - Toxicity prior Beta(0.15, 0.85)	0.3	3.1	1.6	1.4	0.5	
U2 - Toxicity prior Beta(0.3, 0.7)	0.3	3.1	1.6	1.4	0.5	
U3 - Toxicity prior Beta(0.3, 0.7)	0.3	3.1	1.6	1.4	0.5	
U2 - Toxicity prior Beta(0.5, 0.5)	0.3	3.1	1.7	1.4	0.5	
U3 - Toxicity prior Beta(0.5, 0.5)	0.3	3.1	1.7	1.4	0.5	
U2 - Toxicity prior Beta(1, 1)	0.3	3.1	1.7	1.4	0.5	
U3 - Toxicity prior Beta(1, 1)	0.3	3	1.7	1.4	0.5	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	20.6	44.5	12.8	5	1.1	16
U3 - Toxicity prior Beta(0.15, 0.85)	27.7	41.2	10.5	3.9	0.9	15.8
U2 - Toxicity prior Beta(0.3, 0.7)	18.2	45.1	13.4	6.4	1.2	15.7
U3 - Toxicity prior Beta(0.3, 0.7)	24.6	43.4	10.7	5.1	0.7	15.5
U2 - Toxicity prior Beta(0.5, 0.5)	19.5	40.6	16	7.7	1.4	14.8
U3 - Toxicity prior Beta(0.5, 0.5)	25	40	12.9	6.5	1	14.6
U2 - Toxicity prior Beta(1, 1)	19.2	40.8	15.7	8	1.5	14.8
U3 - Toxicity prior Beta(1, 1)	24.6	40.3	13.6	6.1	0.9	14.5
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	10.3	8.7	5.3	2.5	0.9	
U3 - Toxicity prior Beta(0.15, 0.85)	10.3	8.6	5.4	2.5	1	
U2 - Toxicity prior Beta(0.3, 0.7)	10	8.4	5.5	2.6	1.1	
U3 - Toxicity prior Beta(0.3, 0.7)	10.1	8.4	5.4	2.7	1.1	
U2 - Toxicity prior Beta(0.5, 0.5)	9.4	8.6	5.5	3.2	1.3	
U3 - Toxicity prior Beta(0.5, 0.5)	9.3	8.6	5.5	3.2	1.3	
U2 - Toxicity prior Beta(1, 1)	9.3	8.6	5.5	3.2	1.4	
U3 - Toxicity prior Beta(1, 1)	9.2	8.6	5.6	3.2	1.3	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.4	3.6	1.1	0.8	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.4	3.6	1.1	0.8	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.4	3.5	1.2	0.8	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.5	3.5	1.1	0.8	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.5	3.3	1.3	0.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.5	3.3	1.4	0.9	0.2	
U2 - Toxicity prior Beta(1, 1)	0.5	3.2	1.3	0.9	0.2	
U3 - Toxicity prior Beta(1, 1)	0.5	3.3	1.3	0.9	0.2	

Phase I/II Operating Characteristics

Toxicity prior comparison - allocation = two_stage - IPDE alpha = 0.3 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Toxicity prior Beta(0.15, 0.85)	24.9	54	11.2	2.2	0.3	7.4
U3 - Toxicity prior Beta(0.15, 0.85)	30.9	51.5	8.4	1.7	0.2	7.3
U2 - Toxicity prior Beta(0.3, 0.7)	24.7	52.6	12.8	2.5	0.3	7.1
U3 - Toxicity prior Beta(0.3, 0.7)	30.8	51.1	8.7	2.2	0.2	7
U2 - Toxicity prior Beta(0.5, 0.5)	23.4	51.2	14.3	3.9	0.5	6.7
U3 - Toxicity prior Beta(0.5, 0.5)	30.1	48.3	12.1	2.8	0.2	6.5
U2 - Toxicity prior Beta(1, 1)	23.2	51.6	13.9	4.1	0.7	6.5
U3 - Toxicity prior Beta(1, 1)	30.2	49.3	11.2	2.3	0.3	6.7
Mean patients treated						
U2 - Toxicity prior Beta(0.15, 0.85)	9.2	10.9	6	2	0.6	
U3 - Toxicity prior Beta(0.15, 0.85)	9.2	10.9	5.9	2	0.7	
U2 - Toxicity prior Beta(0.3, 0.7)	9.1	10.6	6.1	2.2	0.7	
U3 - Toxicity prior Beta(0.3, 0.7)	9.1	10.5	6	2.2	0.8	
U2 - Toxicity prior Beta(0.5, 0.5)	8.3	10.8	6.1	2.8	0.9	
U3 - Toxicity prior Beta(0.5, 0.5)	8.4	10.7	6	2.8	0.9	
U2 - Toxicity prior Beta(1, 1)	8.4	10.7	6	2.8	0.9	
U3 - Toxicity prior Beta(1, 1)	8.4	10.7	6.1	2.8	0.9	
Mean IPDE patients						
U2 - Toxicity prior Beta(0.15, 0.85)	0.6	4.1	1.2	0.8	0.2	
U3 - Toxicity prior Beta(0.15, 0.85)	0.6	4.1	1.2	0.8	0.2	
U2 - Toxicity prior Beta(0.3, 0.7)	0.6	4	1.3	0.8	0.2	
U3 - Toxicity prior Beta(0.3, 0.7)	0.6	4	1.3	0.8	0.2	
U2 - Toxicity prior Beta(0.5, 0.5)	0.7	3.7	1.5	0.9	0.2	
U3 - Toxicity prior Beta(0.5, 0.5)	0.7	3.7	1.5	0.9	0.2	
U2 - Toxicity prior Beta(1, 1)	0.7	3.7	1.5	0.9	0.2	
U3 - Toxicity prior Beta(1, 1)	0.7	3.7	1.5	0.9	0.2	